A Lightweight Machine Learning Approach for Early Health Risk Prediction

Student Research Proposal • Md Omar Hossain • Machine Learning • Data Science • Healthcare Analytics

Proposed by	Md Omar Hossain
Project type	Student-led research / prototype study
Estimated duration	8–10 weeks

Project Overview

This student project investigates how common machine-learning methods can be used to estimate risk categories from a structured health dataset. The project is intended as an educational research study rather than a clinical diagnostic system. The emphasis will be on responsible data handling, model comparison, interpretability, and honest evaluation.

Student perspective
The project is intentionally scoped so that a small student team can learn, experiment, and produce a useful research artifact without assuming access to a large laboratory or proprietary dataset.

Research Problem

Many healthcare datasets contain useful patterns but also include missing values, class imbalance, correlated features, and other issues that can make a model appear better than it really is. A student-friendly experiment can examine how preprocessing and model choice affect performance and whether a simple model can provide a competitive baseline.

Research Objectives

- Select a suitable public, non-identifiable health dataset with an appropriate research license.
- Study data quality, missing values, class distribution, and feature relationships.
- Build interpretable baseline models such as logistic regression and decision trees.
- Compare the baseline with a stronger ensemble model and evaluate performance using appropriate metrics.
- Use explainability techniques or feature importance to understand the main factors used by the model.

Research Questions

RQ1. How do preprocessing choices affect predictive performance?

RQ2. Which model provides the best balance between performance and interpretability?

RQ3. Which features appear most influential in the selected dataset?

RQ4. How can data leakage and misleading evaluation be avoided?

Proposed Methodology

Step 1. Perform exploratory data analysis and define a reproducible preprocessing pipeline.

Step 2. Split data carefully into training and test sets, avoiding data leakage.

Step 3. Handle missing values and class imbalance using methods appropriate to the dataset.

Step 4. Train several models and compare accuracy, precision, recall, F1-score, and ROC-AUC where appropriate.

Step 5. Perform error analysis and discuss bias, generalization, and why model predictions should not be treated as medical advice.

Data / Materials / Experimental Resources

The exact resources will be finalized with the collaborator after a feasibility discussion. The preferred approach is to start with public or self-generated material, keep the experiment small enough to reproduce locally, and record the source, preprocessing decisions, assumptions, and version of each important component.

Initial resource plan: Public non-identifiable tabular dataset, preprocessing pipeline, train/test split, evaluation notebook.

Evaluation Plan

Evaluation will be based on measurable criteria rather than only a successful demo. Depending on the project, this may include classification metrics, detection quality, latency, error analysis, reproducibility, or qualitative review. Baselines will be established before more complex techniques are introduced so that the contribution of each change can be discussed clearly.

Reproducibility and Documentation

The team will keep a simple experiment log containing dataset versions, preprocessing choices, model or system settings, evaluation results, and important observations. Code will be organized so that the main experiment can be rerun without relying on undocumented manual steps. Charts and tables will be generated from saved results where possible.

Expected Challenges

- Limited time and computing resources may require a smaller experimental scope.
- Dataset quality may affect conclusions more than model complexity.
- A strong-looking metric may hide weaknesses that appear during error analysis.
- The research question may need refinement after an initial feasibility experiment.

Planned Mitigation

The project will begin with a small baseline and a feasibility check. If data or hardware limitations make the original scope unrealistic, the team will narrow the research question rather than artificially inflating results. Each limitation will be reported as part of the final research discussion.

Expected Outcomes

- A reproducible ML pipeline for the selected dataset.
- An evidence-based comparison of several models.
- Feature-importance or explainability analysis.
- A research report emphasizing limitations and responsible interpretation.

Potential Research Collaboration

I am looking for a collaborator interested in machine learning or health-data analysis who can help with experimental design, feature engineering, and interpretation of results. Guidance on responsible use of healthcare data would be particularly valuable.

Technical Stack

Python, pandas, NumPy, scikit-learn, matplotlib, Jupyter Notebook.

Scope, Ethics, and Responsible Research

The study will use non-identifiable data. Results will be reported with uncertainty and limitations. Predictions will not be presented as medical advice or as a replacement for professional diagnosis.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, feasibility check
2. Setup	Dataset/system preparation and baseline implementation
3. Experiments	Development, comparison, tuning, and documentation
4. Evaluation	Metrics, error analysis, limitations, reproducibility
5. Reporting	Results, discussion, documentation, and presentation

What I Am Looking for in a Collaborator

I am looking for someone who is genuinely interested in the research question and willing to contribute to discussion, implementation, experiment design, data analysis, or writing. I would prefer to divide work according to each person's strengths and keep decisions, code, data, and results documented.

Possible Next Step

If a researcher is interested, I would like to first discuss the research question, available resources, and a realistic scope. After that discussion, the proposal can be refined into a clearer study plan with specific responsibilities and milestones.

Note: This document describes a proposed student project. It does not claim completed research, published results, or guaranteed performance. The scope is open to refinement through collaboration.